STANDARD OPERATING PROCEDURES

Bug Reporting, Severity Triaging & Defect Resolution SOP

Issue Lifecycle, Severity SLA Matrix, Jira Template, and QA Verification

Classification: Internal **Version:** 3.1 **Owner:** Quality Assurance & Engineering **Review Cycle:** Annual

1. Objective and Lifecycle

This SOP establishes the standard procedures for identifying, documenting, prioritizing, and resolving software defects across Apex Global Technologies. All defects are tracked in Jira under the standardized lifecycle: `New -> Triaged -> In Progress -> Code Review -> QA Verified -> Closed`.

2. Bug Severity Classification and SLA Matrix

Priority	Impact Description	Initial Response SLA	Target Resolution SLA
P0 (Blocker)	Complete service outage, data corruption, active security breach.	< 15 Minutes (24/7)	< 4 Hours
P1 (Critical)	Core feature broken with no available workaround (e.g., checkout failure, query crash).	< 1 Hour	< 24 Hours
P2 (Major)	Significant functionality impaired, but a documented workaround exists.	< 4 Hours	< 3 Business Days
P3 (Minor)	Minor cosmetic flaw, UI glitch, non-blocking edge-case issue.	< 1 Business Day	Next Scheduled Sprint
P4 (Trivial)	Typo in documentation, minor alignment inconsistency.	< 3 Business Days	Backlog / Opportunistic

3. Mandatory Bug Report Template

All Jira bug tickets must include the following standardized fields:

```
* **Summary:** [Service/Component] Concise description of the failure
* **Environment:** Production / Staging / Dev (Include OS, Browser, App Version)
* **Pre-conditions:** Prerequisites needed to trigger the bug (e.g., User with Editor role)
* **Steps to Reproduce:**
  1. Navigate to `/api/v1/query` endpoint
  2. Post JSON payload with null `collections` array
  3. Observe response status code
* **Expected Result:** API returns HTTP 422 Unprocessable Entity with error envelope
* **Actual Result:** API crashes with HTTP 500 Internal Server Error
* **Logs / Screenshots:** Attach Datadog APM trace link, request_id, and console output
* **Estimated Impact:** Number of affected users or transaction volume
```